

Multiobjective Optimization

A-posteriori Methods: Concepts and Solution Generation

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What are A-posteriori Methods?

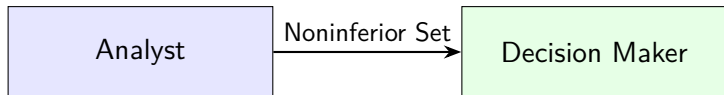
- Methods that **first generate** the noninferior set (or an approximation)
- **Preferences are articulated after** the solution set is presented
- Also called **generating techniques**

Key Philosophy:

"Analysts should analyze, decision makers should decide"

- Decision makers see the **full range of choice**
- Value judgments are made **by decision makers**, not analysts

The Role of Generating Techniques



Generates Multiple Solutions

Chooses Best-Compromise

Bottom-up information flow

- Analyst provides **information**
- Decision maker **chooses** best-compromise
- Preferences are **implicit** in the choice

Why Use A-posteriori Methods?

① Appropriate Roles

- Analysts don't make controversial value judgments
- Decision makers maintain control

② Wider Range of Alternatives

- Multiple options presented, not just one "optimal"
- Informed decision making

③ More Realistic Models

- Real-world problems are multiobjective
- No artificial combination of incommensurable effects

Noninferiority (Pareto Optimality)

*A feasible solution is **noninferior** if no other feasible solution improves one objective without degrading at least one other.*

Best-Compromise Solution

- The noninferior solution selected by decision makers

Tradeoff

- Amount of one objective sacrificed to gain another

GENERATING METHODS

Weighting Method

Constraint Method

Multiobj. Simplex

NISE Method

Key Characteristics:

- No preferences required during solution
- Generate set of alternatives
- Decision makers choose afterwards

Weighting Objectives

Original Multiobjective Problem:

$$\text{maximize } \mathbf{Z} = [Z_1, Z_2]$$

With Weights:

$$\text{maximize } Z(w) = w_1 Z_1 + w_2 Z_2$$

Geometric Interpretation:

- Linear indifference curves
- Slope = $-\frac{w_1}{w_2}$
- Varying weights sweeps across noninferior set

Key Point: Only **relative weights** matter (w_1/w_2)

Weighted Problem:

$$\text{maximize } Z(\mathbf{x}, \mathbf{w}) = \sum_{k=1}^p w_k Z_k(\mathbf{x})$$

Subject to:

$$\mathbf{x} \in \mathbf{F}_d$$

Key Results:

- 1 If $w_k \geq 0$ for all k , solution is **noninferior**
- 2 Only **relative weights** matter (w_i/w_j)
- 3 **Switching values**: ranges of weights giving same solution

Note: If $w_k = 0$ for some k , alternative optima may be inferior

Two-Objective Example

Objective Functions:

$$Z_1(x_1, x_2) = 5x_1 - 2x_2$$

$$Z_2(x_1, x_2) = -x_1 + 4x_2$$

Constraints:

$$-x_1 + x_2 \leq 3$$

$$x_1 + x_2 \leq 8$$

$$x_1 \leq 6, \quad x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Goal: Find noninferior solutions

Weighted Objective:

$$Z(x_1, x_2, w_1, w_2) = w_1(5x_1 - 2x_2) + w_2(-x_1 + 4x_2)$$

Set $w_1 = 1$, vary w_2 :

w_2	Solution	Z_1	Z_2
0	B	30	-6
1	C	26	2
2	D	12	12
3	D	12	12

Instead of weighting objectives, constrain them:

$$\begin{aligned} &\text{maximize } Z_1(\mathbf{x}) \\ &\text{subject to: } \mathbf{x} \in \mathbf{F}_d \\ &Z_2(\mathbf{x}) \geq L_2 \end{aligned}$$

In words:

- 1 Maximize one objective
- 2 Force others to be at least some level
- 3 Vary the levels to trace out noninferior set

Step-by-Step Procedure

Step 1: Construct Payoff Table

- Optimize each objective individually
- Find M_k (max) and n_k (min) for each objective

Step 2: Formulate Constrained Problem

$$\text{maximize } Z_h(\mathbf{x})$$

$$\text{s.t. } \mathbf{x} \in \mathbf{F}_d$$

$$Z_k(\mathbf{x}) \geq L_k, \quad \forall k \neq h$$

Step 3: Choose r values for each L_k

Step 4: Solve for all combinations

Sample Problem Payoff Table

Individual Optimizations:

Solution	x_1	Z_1	Z_2
x^1 (max Z_1)	(6, 0)	30	-6
x^2 (max Z_2)	(0, 4)	-3	15

Ranges:

$$M_1 = 30, \quad n_1 = -6$$

$$M_2 = 15, \quad n_2 = -6$$

Payoff Table:

	Z_1	Z_2
Max Z_1	30	-6
Max Z_2	-3	15

Note: These endpoints define the noninferior set bounds

Sample Application

Constrained Problem:

$$\text{maximize } Z_1 = 5x_1 - 2x_2$$

$$\text{s.t. } \mathbf{x} \in \mathbf{F}_d$$

$$Z_2 = -x_1 + 4x_2 \geq L_2$$

With $r = 4$:

$$L_2 = -6, \quad 1, \quad 8, \quad 15$$

L_2	Solution	Z_1	Z_2
-6	B	30	-6
1	(6, 1.75)	26.5	1
8	(4.8, 3.2)	17.6	8
15	E	-3	15

Aspect	Weighting Method	Constraint Method
Problem changes	Objective function	Constraints
Feasible region	Unchanged	Modified
Nonconvex sets	May miss points	Can find all
Control	Limited	Complete
Infeasible problems	Never	Sometimes

Key Insight:

- Weighting: w_k are weights
- Constraint: L_k are bounds
- Dual variables of constrained problem = weights

Constraint Method Dual Variables:

- In constraint method, the shadow price of $Z_k \geq L_k$ is w_k
- This is the **tradeoff** between objectives
- Equal to weight in weighting method

Theorem: If L_k is binding and shadow price is w_k , then weighting method with w_k gives the **same** noninferior solution.

This demonstrates the **duality** between the two methods.

Problem: Some alternative optima may be inferior

For Weighting Method:

- If some $w_k = 0$, solutions may be inferior
- Check by solving secondary problem
- Maximize objectives with positive weights among the alternatives

For Constraint Method:

- If some constraints are nonbinding, solutions may be inferior
- Check noninferiority using subproblem
- Equivalent to zero-weight case

Key Takeaways

A-posteriori Methods:

- Generate alternatives **before** eliciting preferences
- Decision makers choose best-compromise
- Two main methods: **Weighting** and **Constraint**

Weighting Method:

- Transforms problem to single objective
- Simple, but may miss points in nonconvex cases

Constraint Method:

- Optimizes one objective, constrains others
- More control, but can have infeasible formulations

Topics for Next Lecture

More Generating Methods:

- 1 NISE Method
- 2 Multiobjective Simplex

Displaying Results:

- 1 Value paths
- 2 Projections

Practical Considerations:

- 1 Computational costs
- 2 Choosing appropriate methods



COHON, J. L. *Multiobjective Programming and Planning*. Dover Publications, 2013.